Prismatic cohomology

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1 Grothendieck's mysterious functor problem

Let X be a smooth projective variety over $\mathbb{Q}_p$. Then we have two canonical $\mathbb{Q}_p$-vector spaces:

$$H^i_{et}(X_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_p) \qquad \text{and} \qquad H^i_{dR}(X/\mathbb{Q}_p).$$

Question 1 (Grothendieck, 1971)**.** *Is there a canonical way to relate these two functors? That is, a canonical operation relating the two kinds of cohomology?*

Remark 2. It was mysterious because Grothendieck did not know how to define it but assumed it must exist.

2 Cohomology of complex varieties

Let X be a smooth projective complex variety.

Theorem 3. *There are canonical isomorphisms*

$$H^i_{sing}(X(\mathbb{C}), \mathbb{C}) \xrightarrow{\sim} H^i_{sheaf}(X(\mathbb{C}), \mathbb{C}) \xrightarrow{\sim} H^i_{dR}(X(\mathbb{C})) \xrightarrow{\sim} H^i_{dR}(X).$$

Proof. The first is due to Grothendieck and holds more generally for any abelian group of coefficients and any locally contractible paracompact topological space, the second by the holomorphic Poincaré lemma (which says that $\mathbb{C} \to \Omega^\bullet_{X(\mathbb{C})}$ is a resolution of $\mathbb{C}$), and the last by Grothendieck's theorem [9]. □

Remark 4. There is a canonical $\mathbb{Q}$-structure on the left and a canonical filtration on the right-hand side. The interaction of these two is what leads to *Hodge theory*.

Remark 5. Suppose that X is a smooth projective variety over $\mathbb{Q}$. Then, by the above, we obtain a canonical isomorphism

$$H^i_{sing}(X(\mathbb{C}),\mathbb{Q})\otimes_{\mathbb{Q}}\mathbb{C}\xrightarrow{\sim}H^i_{dR}(X/\mathbb{Q})\otimes_{\mathbb{Q}}\mathbb{C}.$$

This isomorphism is not defined over $\mathbb{Q}$ in general! One really needs to use $\mathbb{C}$ as coefficients.

3 p-adic Hodge theory

Let X be a smooth projective variety over $\mathbb{Q}_p$. Then we have two canonical $\mathbb{Q}_p$-vector spaces:

$$H^i_{et}(X_{\overline{\mathbb{Q}}_p},\mathbb{Q}_p)\qquad\text{and}\qquad H^i_{dR}(X/\mathbb{Q}_p).$$

Remark 6. We have

$$\dim H^i_{et}(X_{\overline{\mathbb{Q}}_p},\mathbb{Q}_p)=\dim H^i_{dR}(X/\mathbb{Q}_p).$$

Indeed, this follows from Theorem 3, by choosing an embedding $\overline{\mathbb{Q}}_p\hookrightarrow\mathbb{C}$.

Remark 7. A canonical isomorphism would provide $H^i_{et}(X_{\overline{\mathbb{Q}}_p},\mathbb{Q}_p)$ with a filtration and $H^i_{dR}(X/\mathbb{Q}_p)$ with an action of $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$, which seems unlikely.

Choose a primitive p^n-th root of unity $\mu_n\in\overline{\mathbb{Q}}_p$ for each $n\geq 1$, set $\mu_0=1$, and suppose that $\mu_n^p=\mu_{n-1}$. Then $K_{cycl}\coloneqq\bigcup_{n\geq 1}\mathbb{Q}_p(\mu_n)$ is Galois, with Galois group $\mathrm{Gal}(K_{cycl}/\mathbb{Q}_p)\cong\mathbb{Z}_p^*$. We get a canonical homomorphism $\chi\colon\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)\to\mathrm{Gal}(K_{cycl}/\mathbb{Q}_p)\cong\mathbb{Z}_p^*$ (this does not depend on the choice of the roots). Let $\mathbb{Z}_p(1)$ be the rank-one $\mathbb{Z}_p$-module equipped with this action of $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$. Let $C=\mathbb{C}_p$ be the completion of $\overline{\mathbb{Q}}_p$; define $C(1)=C\otimes_{\mathbb{Z}_p}\mathbb{Z}_p(1)$ and $C(a)=C(1)^{\otimes_C a}$ for $a\geq 0$, and then $C(a)=\mathrm{Hom}_C(C(-a),C)$ for $a\leq 0$.

Theorem 8 (Faltings, 1988)**.** *Let $C=\mathbb{C}_p$. There is a canonical isomorphism*

$$H^i_{et}(X_{\overline{\mathbb{Q}}_p},\mathbb{Q}_p)\otimes_{\mathbb{Q}_p}C\xrightarrow{\sim}\bigoplus_{a+b=i}H^b(X,\Omega^a_X)\otimes_{\mathbb{Q}_p}C(-a).\tag{1}$$

Proof. See [7]. □

One would like to extend this isomorphism to an isomorphism of filtered vector spaces, so that (1) follows by taking the associated gradeds.

Theorem 9 (Faltings (1989), Tsuji (1999); (C_{dR})-conjecture by Fontaine–Jannsen)**.** *For the* ring of p-adic periods B_{dR}*, there is a canonical isomorphism*

$$H^i_{et}(X_{\overline{\mathbb{Q}}_p},\mathbb{Q}_p)\otimes_{\mathbb{Q}_p}B_{dR}\xrightarrow{\sim}H^i_{dR}(X/\mathbb{Q}_p)\otimes_{\mathbb{Q}_p}B_{dR}.\tag{2}$$

Proof. See [6, 12]. □

Remark 10. The idea is that B_{dR} is equipped with a filtration and a Galois action. The isomorphism is compatible with all these structures.

Let now

$$\mathcal{X} \longrightarrow \mathrm{Spec}(\mathbb{Z}_p)$$

be a smooth projective scheme over $\mathbb{Z}_p$. There are two problems:

(a) The isomorphism (2) is not integral. In particular, all torsion information is lost.

(b) There are many more cohomology theories attached to our situation:

   (a) Étale cohomology of the generic fiber: $H^i_{et}(\mathcal{X}_{\bar{\eta}}, \mathbb{Z}_p)$ and in particular the group $H^i_{et}(\mathcal{X}_{\bar{\eta}}, \mathbb{Z}/p)$.

   (b) The de Rham cohomology of the total space and the generic fiber: $H^i_{dR}(\mathcal{X}/\mathbb{Z}_p)$ and $H^i_{dR}(\mathcal{X}_\eta/\mathbb{Q}_p)$ (the former recovers the latter by inverting p).

   (c) The de Rham cohomology of the special fiber: $H^i_{dR}(\mathcal{X}_0/\mathbb{F}_p)$. Since we have $R\Gamma(\mathcal{X}_{0,cris}, \mathbb{Z}_p) \otimes^L_{\mathbb{Z}_p} \mathbb{F}_p \simeq R\Gamma(\mathcal{X}_0, \Omega^\bullet_{\mathcal{X}_0}) \in D(\mathbb{F}_p)$, we have an exact sequence

$$0 \to H^i_{crys}(\mathcal{X}_0/\mathbb{Z}_p) \otimes_{\mathbb{Z}_p} \mathbb{F}_p \to H^i_{dR}(\mathcal{X}_0/\mathbb{F}_p) \to H^{i+1}_{crys}(\mathcal{X}_0/\mathbb{Z}_p)[p] \to 0.$$

   (d) Note that, in this case, $H^i_{crys}(\mathcal{X}_0/\mathbb{Z}_p) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \cong H^i_{dR}(\mathcal{X}/\mathbb{Z}_p) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ [3, Corollary 3.8]. One could also look at the étale cohomology $H^i_{et}(\mathcal{X}_0, \mathbb{Z}/p)$ of the special fiber.

Question 11. *Is there an integral cohomology theory that recovers all of these groups?*

4 Prismatic cohomology

To solve the above problem, we first need to have some integral period rings, as well as an appropriate site and a sheaf on that site. This leads to the following definition.

Definition 12. A *prism* is a triple (A, ϕ, Z) consisting of a δ-ring A, its associated lift of Frobenius $\phi\colon A \to A$, and a Cartier divisor $Z = Z(I) \subset \mathrm{Spec}(A)$, such that $p \in I + \phi(I)A$; in particular, Z and $\phi^{-1}(Z)$ meet only in characteristic p. One also requires that A is derived (p, I)-complete. A prism is *bounded* if A/I has bounded p^∞-torsion [5, Definitions 1.1 and 1.4].

Remark 13. We actually require a bit more than a lift of Frobenius: we want that A is a δ-ring, so that the lift of Frobenius is defined by $\phi(x) = x^p + p\delta(x)$. Saying that Z and $\phi^{-1}(Z)$ meet only in characteristic p means that $p \in I + \phi(I)A$.

Examples 14. 1. (Breuil–Kisin prism.) Let $A = \mathbb{Z}_p[[u]]$ and let $I = (u - p)$. Here, $\phi|_{\mathbb{Z}_p} = \mathrm{id}$ and $\phi(u) = u^p$. See Figure 1 below.

2. (*Perfect prisms.*) Let C be the p-adic completion of $\overline{\mathbb{Q}}_p$, and let $\mathcal{O}_C \subset C$ be its ring of integers. Define

$$\mathcal{O}_C^\flat = \varprojlim_{x \mapsto x^p} \mathcal{O}_C/p, \qquad C^\flat = \mathrm{Frac}(\mathcal{O}_C^\flat), \qquad A_{\mathrm{inf}} = W(\mathcal{O}_C^\flat).$$

The ring $\mathcal{O}_C^\flat$ is called the *tilt* of $\mathcal{O}_C$, and there is a noncanonical isomorphism $C^\flat \simeq \widehat{\overline{\mathbb{F}_p((t))}}$. The natural map $\varprojlim_{x \mapsto x^p} \mathcal{O}_C \to \mathcal{O}_C^\flat$ is an isomorphism of multiplicative monoids. For $x \in \mathcal{O}_C^\flat$, let $x^\sharp \in \mathcal{O}_C$ be the zeroth component of the corresponding compatible sequence in $\mathcal{O}_C$. Fontaine's ring map $\theta\colon A_{\mathrm{inf}} \to \mathcal{O}_C$ characterized by $\theta([x]) = x^\sharp$. Equivalently, $\theta\left(\sum_{n\geq 0} p^n[x_n]\right) = \sum_{n\geq 0} p^n x_n^\sharp$. Choose a compatible sequence of p-power roots of p, and let $p^\flat \in \mathcal{O}_C^\flat$ be the induced element. Then $\xi = [p^\flat] - p$ is a generator of $\ker(\theta)$, up to multiplication by a unit [8, § 2.1.3]. $\bar{\xi} \in \mathcal{O}_C^\flat$. Thus $(A_{\mathrm{inf}}, (\xi))$ is a perfect prism.

3. Let $A = \mathbb{Z}_p[[u]]$ be as before. The chosen roots determine a morphism of prisms

$$(A, (u-p)) \longrightarrow (A_{\mathrm{inf}}, (\xi)), \qquad u \longmapsto [p^\flat]. \tag{3}$$

This morphism is not canonical: it depends on the chosen compatible roots of p.

4. Define the *crystalline prism* as $(\mathbb{Z}_p, (p))$, equipped with $\phi = \mathrm{id}$.

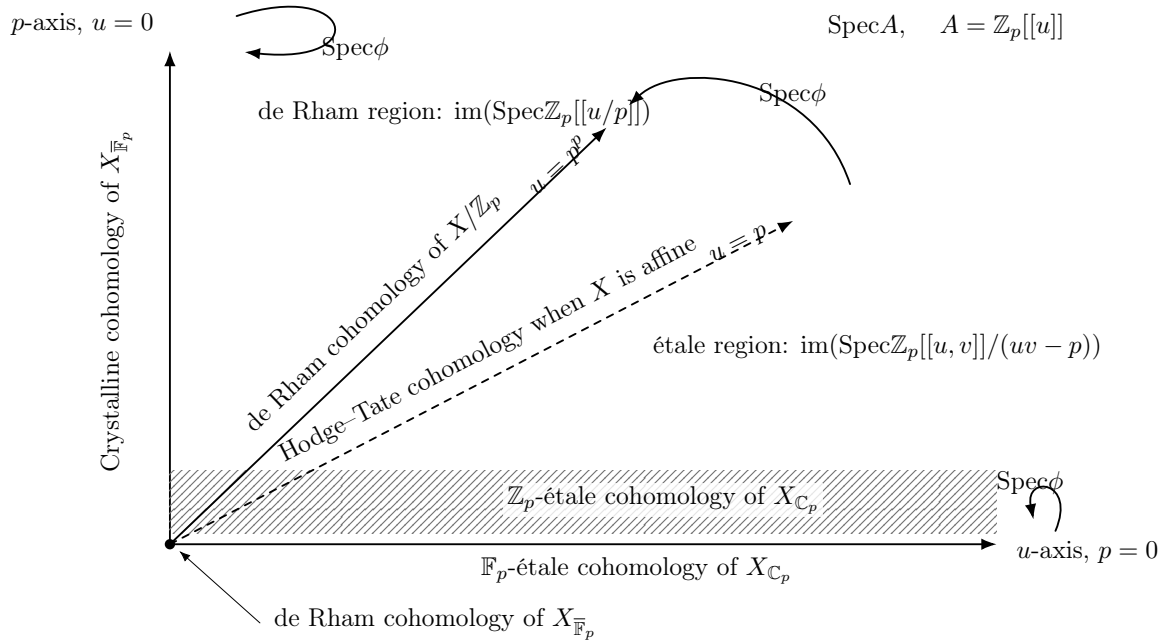


Figure 1: The specializations of $R\Gamma_\Delta(X/A)$ at various points and regions of Spec$A$ for $A = \mathbb{Z}_p[[u]]$. Figure taken from [10, Figure 1.4.6].

Theorem 15. *(See [5, Theorem 1.8].) For each bounded prism (A, I) and each smooth projective morphism $X \to \mathrm{Spec}(A/I)$, write X also for its p-adic formal completion. There exists a canonical object*

$$R\Gamma_\Delta(X/A) \in D(A) \tag{4}$$

in the derived category of A-modules, such that the following holds:

1. *(de Rham comparison for the Breuil–Kisin prism.) Let $A = \mathbb{Z}_p[[u]]$ be the Breuil–Kisin prism, see item (1) of Examples 14. There is a canonical isomorphism*

$$R\Gamma_\Delta(X/A)\widehat{\otimes}^L_{A,u\mapsto p^p}\mathbb{Z}_p \xrightarrow{\sim} R\Gamma_{dR}(X/\mathbb{Z}_p) \qquad in \qquad D(\mathbb{Z}_p).$$

2. *(de Rham comparison for A_{inf}.) Similarly, if $A_{inf} = W(\mathcal{O}_C^\flat)$ is the above-defined perfect prism, see item (2) of Examples 14, so that $X \to \mathrm{Spec}(\mathcal{O}_C)$ is a smooth projective scheme over $\mathcal{O}_C = A_{inf}/(\xi)$, there is a canonical isomorphism*

$$R\Gamma_\Delta(X/A_{inf})\widehat{\otimes}^L_{A_{inf},\theta\circ\phi}\mathcal{O}_C \xrightarrow{\sim} R\Gamma_{dR}(X/\mathcal{O}_C) \qquad in \qquad D(\mathcal{O}_C).$$

3. *(Crystalline comparison) Let $A = \mathbb{Z}_p[[u]]$ be the Breuil–Kisin prism, see item (1) of Examples 14. We have a canonical isomorphism*

$$R\Gamma_\Delta(X/A)\widehat{\otimes}^L_{A,u\mapsto 0}\mathbb{Z}_p \xrightarrow{\sim} R\Gamma_{crys}(X_0/\mathbb{Z}_p) \qquad in \qquad D(\mathbb{Z}_p).$$

4. *(Étale comparison.) Let $A_{inf} = W(\mathcal{O}_C^\flat)$ be the above-defined perfect prism, see item (2) of Examples 14. We have a canonical isomorphism*

$$\left(R\Gamma_\Delta(X_{\mathcal{O}_C}/A_{inf}) \otimes^L_{A_{inf}} C^\flat\right)^{\phi=1} \xrightarrow{\sim} R\Gamma_{et}(X_C, \mathbb{Z}/p) \qquad in \qquad D(\mathbb{Z}/p),$$

   where $(\phi = 1)$ denotes derived Frobenius fixed points [5, Theorem 9.1], or, equivalently,

$$R\Gamma_\Delta(X_{\mathcal{O}_C}/A_{\mathrm{inf}}) \otimes^L_{A_{\mathrm{inf}}} C^\flat \simeq R\Gamma_{\text{ét}}(X_C, \mathbb{F}_p) \otimes_{\mathbb{F}_p} C^\flat \qquad in \qquad D(C^\flat),$$

   where the linearization follows from [2, Lemma 22].

5. *(Base change.) Consider a morphism of bounded prisms $(A, I) \to (B, J)$ (see (3) for an example; another example is given by the map $(\mathbb{Z}_p[[u]], (u-p)) \to (\mathbb{Z}_p, (p))$ which is reduction modulo u). There is a canonical isomorphism*

$$R\Gamma_\Delta(X/A)\widehat{\otimes}^L_A B \xrightarrow{\sim} R\Gamma_\Delta(X_{B/J}/B) \qquad in \qquad D(B).$$

Remark 16. With regard to the first and third items in Theorem 15, we have in fact a canonical isomorphism $R\Gamma_{dR}(X/\mathbb{Z}_p) \cong R\Gamma_{crys}(X_0/\mathbb{Z}_p)$ [3, Corollary 3.8]. This is due to the fact that we took a smooth projective lift over $\mathbb{Z}_p = W(\mathbb{F}_p)$, rather than over the ring of integers of a ramified finite extension of $\mathbb{Q}_p$. (Clearly, it would be more natural to phrase everything for finite extensions of $\mathbb{Q}_p$; we refrained from this for expositional reasons.) This isomorphism does not hold in this form when we replace $\mathbb{Q}_p$ by a finite extension with large ramification index.

The following is a rather straightforward corollary of Theorem 15. To phrase it, we will use the following:

Definition 17. For any bounded prism (A, I), define

$$R\Gamma_{\Delta/p}(X/A) \coloneqq R\Gamma_{\Delta}(X/A) \otimes^{L}_{A} (A/pA).$$

We then have the following result, which is a rather direct consequence of Theorem 15, and appears to be useful in applications.

Theorem 18. *Let $A_{inf} = W(\mathcal{O}^{\flat}_{C})$ with its prism structure defined by the ideal $(\xi) = \mathrm{Ker}(\theta)$, see Example 14(2) above. Define a functor*

$$X \longmapsto H^{i}_{\Delta/p}(X/A_{inf}) \coloneqq H^{i}\left(R\Gamma_{\Delta/p}(X/A_{inf})\right)$$

from the category of smooth projective schemes $X \to \mathrm{Spec}(\mathcal{O}_C)$ to the category of A_{inf}-modules. This functor has the following properties.

1. *(de Rham comparison.) View $\overline{\mathbb{F}}_p$ as an $\mathcal{O}^{\flat}_{C}$-module via $\mathcal{O}^{\flat}_{C} \xrightarrow{\phi} \mathcal{O}^{\flat}_{C} \to \overline{\mathbb{F}}_p$, in which the second map is the residue map. Then we have a canonical exact sequence*

$$0 \to H^{i}_{\Delta/p}(X/A_{inf}) \otimes_{\mathcal{O}^{\flat}_{C},\phi} \overline{\mathbb{F}}_p \to H^{i}_{dR}(X_{\overline{\mathbb{F}}_p}) \to \mathrm{Tor}_{1}^{\mathcal{O}^{\flat}_{C}}\left(H^{i+1}_{\Delta/p}(X/A_{inf}), \overline{\mathbb{F}}_p\right) \to 0.$$

   See [8, § 2.1.5(1)].

2. *(Étale comparison.) We have a canonical isomorphism,*

$$H^{i}_{et}(X_C, \mathbb{F}_p) \otimes_{\mathbb{F}_p} C^{\flat} \xrightarrow{\sim} H^{i}_{\Delta/p}(X/A_{inf}) \otimes_{\mathcal{O}^{\flat}_{C}} C^{\flat}.$$

   See [8, Lemma 2.1.6].

5 Construction of the site and the sheaf

Let us now explain the key behind the definition of the object $R\Gamma_{\Delta}(X/A)$ of (4).

Definition 19 (Definition 1.6)**.** Fix a bounded prism (A, I) and a smooth p-adic formal scheme X over A/I [5, Definition 1.6].

1. A map $(A, I) \to (B, J)$ of prisms is called *(faithfully) flat* if $A \to B$ is (p, I)-completely (faithfully) flat.

2. The *prismatic site*

$$(X/A)_{\Delta}$$

   is the opposite of the category of prisms (B, J) equipped with a map $(A, I) \to (B, J)$ and a map $\mathrm{Spf}(B/J) \to X$ over $\mathrm{Spf}(A/I)$, endowed with the faithfully flat covers defined in (1).

3. The *structure sheaf* $\mathcal{O}_{\Delta}$ on $(X/A)_{\Delta}$ is the sheaf which sends a pair (B, J), together with maps $(A, I) \to (B, J)$ and $\mathrm{Spf}(B/J) \to X$, to B. This is a sheaf of I-torsionfree δ-A-algebras.

4. There is also another sheaf of rings $\overline{\mathcal{O}}_{\Delta}$ on $(X/A)_{\Delta}$ which sends a pair (B, J) to B/J. This is naturally a sheaf of $\mathcal{O}(X)$-algebras, and we have $\mathcal{O}_{\Delta} \otimes_{A} A/I \xrightarrow{\sim} \overline{\mathcal{O}}_{\Delta}$.

One then defines

$$R\Gamma_{\Delta}(X/A) \coloneqq R\Gamma((X/A)_{\Delta}, \mathcal{O}_{\Delta}).$$

6 Applications

We give some applications.

Corollary 20 (Bhatt–Morrow–Scholze)**.** *Let X be a smooth projective variety over $\mathbb{Q}$ with good reduction at a prime p. Then*

$$\dim H^i_{sing}(X(\mathbb{C}), \mathbb{Z}/p) \leq \dim H^i_{dR}(X_p).$$

See [4, Theorem 1.1(ii)].

Proof. Choose a smooth projective model over $\mathbb{Z}_p$, and continue to denote its base change to $\mathcal{O}_C$ by X. We have $\dim H^i_{sing}(X(\mathbb{C}), \mathbb{Z}/p) =$

$$\begin{aligned}\dim \left(H^i_{et}(X_C, \mathbb{F}_p)\right) &= \dim \left(H^i_{et}(X_C, \mathbb{F}_p) \otimes_{\mathbb{F}_p} C^\flat\right) = \dim \left(H^i_{\Delta/p}(X/A_{inf}) \otimes_{\mathcal{O}_C^\flat} C^\flat\right) \\ &\leq \dim \left(H^i_{\Delta/p}(X/A_{inf}) \otimes_{\mathcal{O}_C^\flat, \phi} \overline{\mathbb{F}}_p\right) \leq \dim \left(H^i_{dR}(X_{\overline{\mathbb{F}}_p})\right).\end{aligned}$$

This proves what we want. □

Corollary 21 (Bhatt–Morrow–Scholze)**.** *There exists a smooth projective geometrically connected (relative) surface $X \to \mathrm{Spec}(\mathbb{Z}_2)$ such that:*

(i) the étale cohomology groups $H^i_{\text{ét}}(X_{\overline{\mathbb{Q}}_2}, \mathbb{Z}_2)$ are free over $\mathbb{Z}_2$ for all $i \in \mathbb{Z}$; and

(ii) the second crystalline cohomology group $H^2_{\text{crys}}(X_{\mathbb{F}_2}/\mathbb{Z}_2)$ has nontrivial 2-torsion, namely $H^2_{\text{crys}}(X_{\mathbb{F}_2}/\mathbb{Z}_2)_{\text{tor}} \cong \mathbb{F}_2$ [4, Theorem 2.1].

In particular, there is no abstract isomorphism

$$H^2_{crys}(X_{\mathbb{F}_2}/\mathbb{Z}_2) \cong H^2_{et}(X_{\overline{\mathbb{Q}}_2}, \mathbb{Z}_2).$$

Corollary 22 (Lang, Illusie)**.** *Let Y be an Enriques surface over $\mathbb{F}_2$. (That means that over the algebraic closure, $\kappa(Y_{\bar{\mathbb{F}}_2}) = 0$ and $b_2(Y_{\bar{\mathbb{F}}_2}) = 10$.) Then*

$$H^1_{dR}(Y) \neq 0.$$

Proof. After possibly enlarging the ground field, Y admits a smooth projective lift X over the ring of integers of a finite extension of $\mathbb{Q}_2$ [11, Theorem 4.10]. Choosing an embedding $\overline{\mathbb{Q}}_p \hookrightarrow \mathbb{C}$, the argument above shows that

$$\dim H^1_{sing}(X(\mathbb{C}), \mathbb{Z}/2) \leq \dim H^1_{dR}(Y).$$

Now

$$H^1_{sing}(X(\mathbb{C}), \mathbb{Z}/2) = \mathrm{Hom}(\pi_1(X(\mathbb{C})), \mathbb{Z}/2) = \mathbb{Z}/2.$$

Hence the result follows. □

Theorem 23 (Farb–Kisin–Wolfson, 2024)**.** *Let X be a smooth projective complex variety of maximal Albanese dimension, and let $Y \to X$ be the étale cover defined by the maximal elementary p-quotient $\pi_1(X) \twoheadrightarrow (\mathbb{Z}/p)^n$. Then for $p \gg 0$, the map $Y \to X$ is p-incompressible. That is, for no dense open $U \subset X$ and no finite cover $U' \to U$ of degree coprime to p does the map $Y_{U'} \to U'$ arise as a pull-back of a finite map $Y_0 \to X_0$ with $\dim X_0 < \dim X$*

Proof. See [8, Theorem 1]. □

Remark 24. This applies to abelian varieties, where $Y \to X$ will be the multiplication by p map $[p]\colon X \to X$. This (almost) solves a conjecture of Brosnan, see [8, Remark 2.3.6]. It also gives interesting examples when applying it to products of curves, and certain compact ball quotients.

Theorem 25 (Ambrosi–Newton–Pagano, 2025)**.** *Let X be a smooth projective variety over a number field K. Assume that $H^0(X, \Omega_X^2) \neq 0$. Then X fails weak approximation for some finite extension L/K*

(That is, $X(L) \to \prod_{v \in S} X(L_v)$ is not dense for some finite set of places S of L.)

Proof. See [1, Corollary 1.1.2]. □

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